PURDUE
UNIVERSITY

MGMT 30500 – Sections 2 and 3
Business Statistics – Spring 2026

Instructor: Tongseok Lim
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Class:
Section 2 9:30 am – 10:20 am MWF, WALC 2051
Section 3 3:30 pm – 4:20 pm MWF, WALC 2088

Office Hours (In-person, KRAN 430):

10:40 am – 12:30 pm Monday (or by appointment)

Prerequisite: STAT 30301: Probability and Statistics for Business.

COURSE MATERIALS: The following are learning materials:

Text: Anderson, Sweeney, Williams, Camm, Cochran, Fry, and Ohlmann: *Modern Business Statistics with Microsoft Excel*, 7th edition, 2018, Cengage Learning, Inc. **(Required)** This text is also used in STAT 30301.

Handouts: PowerPoints presentation slides and Supplementary Materials. (Posted on Brightspace: Brightspace -> Content -> Table of Contents -> ….)

COURSE WEBSITE: Purdue Brightspace.

Important announcements and information will be posted on Brightspace. **You should check the course website regularly.**

Please make your own copies of all course materials. There will be no handouts in the classes.

SOFTWARE: The software known as **EXCEL** will be used for in-class demonstrations and instruction. The main tool is Data Analysis under Tools. If you don't see this tool, follow the following steps to add in:

File → Options → Add-ins → Select Analysis ToolPak and Analysis ToolPak-VBA (also select StatTools 7.5, if available) → Go → Data → Data Analysis (to conduct analyses)

Course Objectives:

This course is designed to introduce students to basic data analysis techniques. Coverage will include the application of *probability distributions* such as the normal, the t, and the binomial; *sampling distributions*, *basic statistical inference methods*, *analysis of variance, applied linear regression* techniques, *logistic regression, time series analysis, statistical quality control, and decision analysis*.

Our main goal is to instill the basic quantitative and data analysis skills needed by managers in modern business, where Business Analytics and Data Science have become important. These skills will help managers to understand and carry out data-based decision making, risk assessments, and policy making and to work effectively with data analysis teams to improve all aspects of business performance.

Learning Outcomes:

1. Understand the basic statistical principles and their applications in the area of management and business.
2. Understand fundamental issues in business and management.
3. Describe and use the commonly used statistical techniques for analyzing business data.
4. Be proficient in using Excel to carry out statistical and analytics methods covered in the course.

Usage of Generative Artificial Intelligence/Large Language Models:

Generative AI systems, such as ChatGPT or Gemini, can serve as valuable learning tools when used appropriately. Students are welcome to use these tools as a mentor—to ask questions, clarify concepts, explore examples, and deepen their understanding of course material.
However, generative AI should not be used to directly produce or complete course deliverables.

All course assignments, quizzes, and exams must reflect each student's own independent work and be completed in Microsoft Excel or other tools as instructed. Submitting AI-generated answers, formulas, or solutions from others' work as one's own is considered a violation of academic integrity. Violations can include a failing grade for the course and restrictions from further class attendance. All suspected incidents of academic dishonesty will also be referred to the Office of Student Rights and Responsibilities for further review of the student's status with the University, which may include being separated from the University.

ASSESSMENTS

Grading: Following a university-wide initiative, the School has adopted an Official Grading Policy for core courses such as MGMT 30500. Under this policy, the overall GPA for this class can be no higher than 3.0. To comply with this course policy, final course letter grades will be based on the curved class final course percentages. The final course percentage is based on a weighted percentage computed using the weights shown in the table below:

Homework*	24%
Quizzes**	21%
Midterms***	30%
Attendance/Participation	5%
Final****	20%

*There are eight homework assignments. No homework assignment is dropped. The final homework score is calculated using aggregated points.
** There are three quizzes worth 7% each. Quizzes 2 and 3 are not cumulative.
***There are two midterms: each is worth 15%. Midterms are not cumulative.
**** Final is cumulative.

Homework: There are eight Homework Assignments. Each Homework Assignment contains

- exercise problems from the text, and
- supplementary problems.

Please note: Homework is accepted only by being properly and timely uploaded to Brightspace. Homework is not accepted late or in any other way, including by email, for any reason. Also, all students must be assessed on the same work so ad-hoc extra-credit or makeup assignments will not be given.

Homework assignments must be typed and submitted online through Brightspace. Brightspace will be set to allow you multiple chances to submit the assignment. Only your latest submission will be graded if you submit more than once.

Make sure to upload only one pdf file for each homework assignment. No supplements, attachments, or appendices can be used. Do not upload Word or page documents.

Important: You must include all relevant software output that addresses the homework questions and reference it clearly where appropriate. Copy and paste all software output that addresses the homework questions into your homework papers, and reference it clearly where appropriate, but **do not copy and paste any datasets into your homework papers.**

If an exercise instructs you to do a computation by hand, you do not have to include the by-hand work in your homework paper. This makes it easier to turn your work in online to Brightspace. However, you should make sure that you can do the by-hand computation by checking your answer against Excel's result. You will need to do the by-hand computations on exams so, again, make sure you know how to do them.

In your submissions, homework problems must be clearly identified by their respective chapter and problem numbers.

Your proficiency in this course will improve if these problems are done faithfully and in a timely fashion. **Try not to wait until the last minute to do your homework.**

You may discuss the homework problems with classmates, but DO NOT COPY. You are expected to write your homework individually. **Duplicate papers are considered cheating.**

Your homework submissions are graded by the course graders. If you have any issues about the grading, please refer to the paragraph, Grade Challenges, below.

Quizzes: To encourage students to work steadily through the materials, three **in-class** quizzes are given. Quizzes 2 and 3 are not cumulative. The quizzes will each count for 7% of the final course percentage, and none will be dropped. Only under **exceptional circumstances,** when legitimate and verifiable reasons are provided, will make-ups be given. "Exceptional circumstances" means a death in the family, a serious *personal* medical emergency, participation in a conflicting NCAA athletic event, or as otherwise required by University policies. ***Except for emergencies*, *requests for a makeup must be made by email with supporting documentation no later than 7 days before the scheduled quiz.***

Exams: Exams will test both technical ability to carry out standard calculations and your understanding of important concepts. Exams are closely based on homework exercises**.** There will be two Midterms and a Final, **all given in-person**. The second Midterm is not cumulative. **The Final is cumulative.** You will need a calculator for taking exams. Calculators that have USB ports or can access the internet are not permitted nor are you allowed to use a smart phone or watch as a calculator. You are required to take all exams. (None are dropped.) Only under **exceptional circumstances,** when legitimate and verifiable reasons are provided, will make-ups be given. "Exceptional circumstances" means a death in the family, a serious personal medical emergency, participation in a conflicting NCAA athletic event, or as otherwise required by University policies. ***Except for emergencies, requests for a makeup must be made by email with supporting documentation no later than 7 days before the scheduled exam.***

Attendance and Participation: Attend your classes, pay attention and participate. If you do not attend class, you will likely not succeed. According to the University regulations (https://catalog.purdue.edu/content.php?catoid=15&navoid=18634#classes), **"**Scheduled courses allow students to avoid conflicts and reflect the University's expectation that students should be present for every meeting of a class/laboratory for which they are registered." The instructor will take random attendance and keep the attendance record. Your attendance/participation grade will be based on this record.

To enhance engagement, we will occasionally (and possibly in every class) administer short in-class quizzes in addition to the three officially scheduled quizzes. In these cases, your submitted quiz paper will also be used to record attendance.

Grade Challenges: When grades are posted in the Brightspace Grade Center, an announcement will be posted on Brightspace. Any challenge to any grade must be made **within 7 calendar days** of the posting of the announcement. There is one exception: to facilitate the computation of final course grades, the last homework assignment must be challenged **within three days of the posting of grades.** Grade challenges must be based on good faith disputes about mathematical or statistical principles. They cannot be based on post-hoc legalistic arguments.

To challenge a homework score, quiz score or exam score, you must send the instructor an email describing your challenge**.** Include your name, ID, and Assignment Number. Please note: General grade reviews for homework are not done. For homework, the challenge must cite the instructor to a specific deduction and provide a reason it is believed to be incorrect. Please check the Homework Solutions before submitting a homework grade challenge. Most often, the Homework Solutions will provide the basis for the deduction(s). **Grades will not be discussed in the classroom during, between, or after classes.**

After the 7 day challenge period, all grades are final for purposes of calculating the final course grade.

KEYS TO SUCCESS: You will succeed in this course if you do the following:

- First**, as noted above, work steadily following the schedule in the Course Outline.**
- Second, read the lecture material **before your scheduled class**. (See the Course Schedule to determine what chapter to read.) Usually the readings are fairly short, particularly when accounting for tables and figures.
- **Before class**, make a quick review of the homework exercises for that chapter. These are mainly what we will cover in class.
- Then, either print the readings and the homework exercises, and bring them to class to take notes directly on the reading material. Or, make sure you have these materials readily available on your laptop or tablet to follow the in-class presentation.
- **Problems**. The only way you can really understand statistics and data analysis is by seeing many examples first-hand and doing many problems. It is important to realize that this course is different than many others in the following respect: you may feel that you understand both the lectures and the readings quite well, but then find that, when you first try problems, you are unable to do them. You must try to do the assigned homework problems on your own and work the chapter problems on your own as we go along as much as possible. If you do work regularly (after each chapter lecture is completed), you will quickly learn to do the problems.
- If you follow these procedures, you will succeed in this class. **If you do not read the course materials and just try to absorb the material through the in-class presentations, you will likely not succeed.**

Deadlines: The Registrar's add, drop, and modify deadline dates for this semester are at the following website: http://www.purdue.edu/Registrar/

Academic Integrity: Please refer the University Policies on Brightspace.

Accessibility and Accommodations:

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, please let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

- **CAPS Information**: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

Non-Discrimination Statement: Please refer to the Nondiscrimination Statement on Brightspace.

Emergency Situation(s): Please refer to the Emergency Preparedness on Brightspace.

TENTATIVE COURSE SCHEDULE – Spring 2026

WEEK	DATE	DAY	TOPIC	MATERIALS*	DUE
1	Jan 12	M	Intro. & Basic Stat. & Prob. Rvw.	Syl., 1.2, 1.4, 1.5	
	Jan 14	W	Basic Stat. & Prob. Rvw. (cont'd)	3.3 (omit Chebyshev), 3.4, 3.5	
	Jan 16	F	Basic Stat. & Prob. Rvw. (cont'd)	6.2, t Dist. Supplement	
	Jan 18	Sunday			HW1
2	**Jan 19**	**M**	**MARTIN LUTHER KING JR. DAY**	**No Class**	
	Jan 21	W	Int. Est. Rvw.	8.1, 8.2, 8.3, 8.4	
	Jan 23	F	Hyp. Testing Rvw.	9.1, 9.2, 9.3, 9.4, 9.5	
	Jan 25	Sunday			HW 2
3	Jan 26	M	Hyp. Testing Rvw. (cont'd)	11.2	
	Jan 28	***W***	***Quiz 1 (Diagnostic)***	***Review Material (Chs. 1 – 12)***	
	Jan 30	F	Analysis of Variance	13.1, 13.2	
4	Feb 2	M	Simple Regression	14.1, 14.2	
	Feb 4	W	Simple Regression (cont'd)	14.2, 14.3	
	Feb 6	F	Simple Regression (cont'd)	14.4, 14.5	
5	Feb 9	M	Simple Regression (cont'd)	14.6, 14.7	
	Feb 11	W	Simple Regression (cont'd)	14.8	
	Feb 13	F	Simple Regression (cont'd)	14.8, 14.9	
	Feb 15	Sunday			HW3
6	Feb 16	M	Midterm 1 Review	Analysis of Variance and Simple Regression	
	Feb 18	W	Midterm 1 Review	Analysis of Variance and Simple Regression	
	Feb 19	***Th***	***Midterm 1***		
	Feb 20	F	No class	in lieu of the Midterm Exam	
7	Feb 23	M	Multiple Regression (cont'd)	15.1, 15.2	
	Feb 25	W	Multiple Regression (cont'd)	15.3, 15.5, 15.5	
	Feb 27	F	Multiple Regression (cont'd)	15.7	
8	Mar 2	M	Multiple Regression (cont'd)	15.4, 15.6, 15.8	
	Mar 4	W	Multiple Regression (cont'd)	Applications	
	Mar 6	F	Multiple Regression (cont'd)	Applications	
	Mar 8	Sunday			HW4
9	Mar 9	M	Model Building (cont'd)	16.1, 16.2	
	Mar 11	***W***	***Quiz 2***	***Multiple Regression***	
	Mar 13	F	Model Building (cont'd)	16.3, 16.4	
10	Mar 16	**M**	**Spring Break**	**No Class**	
	Mar 18	**W**			
	Mar 20	**F**			
	Mar 22	Sunday			HW5
11	Mar 23	M	Logistic Regression	Supplement	
	Mar 25	W	Logistic Regression (cont'd)	Supplement	
	Mar 27	F	Logistic Regression (cont'd)	Supplement	
	Mar 29	Sunday			HW6
12	Mar 30	M	Midterm 2 Review	Mult. Regr., Model Bldg, Log. Regr.	
	Apr 1	W	Midterm 2 Review	Mult. Regr., Model Bldg, Log. Regr.	
	Apr 1	***W***	***Midterm 2***	***Mult. Regr., Model Bldg, Log. Regr.***	
	Apr 3	F	No class	in lieu of the Midterm Exam	
13	Apr 6	M	Time Series (cont'd)	17.1, 17.2, 17.3, 17.4	
	Apr 8	W	Time Series (cont'd)	17.5, 17.6	
	Apr 10	F	Time Series (cont'd)	Lag Variables Supplement & Applications	
	Apr 12	Sunday			HW7
14	Apr 13	W	Quality Control	19.1 19.2	
	Apr 15	***W***	***Quiz 3***	***Time Series***	
	Apr 17	F	Quality Control (cont'd)	19.2	
15	Apr 20	M	Quality Control (cont'd)	19.3	
	Apr 22	W	Decision Analysis	20.1, 20.2	
	Apr 24	F	Decision Analysis (cont'd)	20.3, 20.4	
	Apr 26	Sunday			HW8
16	Apr 27	M	Final Exam Review 1	Cumulative	
	Apr 29	W	Final Exam Review 2	Cumulative	
	May 1	**F**	**Final Exam Preparation**	**No Class**	
17	May 4 - 8		***Final Exam***		

***Section Numbers refer to sections in the course textbook:** Modern Business Statistics with Microsoft Excel, 7th Edition**, by Anderson, Sweeney, et al**

HOMEWORK SCHEDULE – Spring 2026

ASSIGNMENT	TOPIC	DUE DATE
1	Basic Stat. & Prob. Rvw.	Jan. 18
2	Int. Est. & Hyp. Testing Rvw.	Jan. 25
3	Simple Regression	Feb. 15
4	Multiple Regression	Mar. 8
5	Model Building	Mar. 22
6	Logistic Regression	Mar. 29
7	Time Series	Apr. 12
8	Quality Control and Decision Analysis	Apr. 26

^**Assignments must be uploaded to Brightspace by 11:59 pm Eastern US on the listed Due Date. Note: all due dates are Sundays.**

To find and submit all homework assignments: Brightspace -> Course Tools -> Assignments.

Reminder: Homework is not accepted late or in any other way for any reason.